Siyuan Dong

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University of Michigan, Ann Arbor Ann Arbor, MI

SUMMARY

I am a CS Ph.D. student at the University of Michigan, Ann Arbor, advised by Prof. Lin Ma. My research lies at the intersection of machine learning and data systems. I currently work on learned query optimization across heterogeneous database systems, with an emphasis on building methods that are accurate, efficient, and reusable across different database engines. I also study automated optimization for lakehouse systems, including the joint selection and configuration of execution engines and table formats. More broadly, I am interested in intelligent and portable systems for data management.

EDUCATION

University of Michigan, Ann Arbor Ann Arbor, MI
Ph.D., Computer Science and Engineering Expected May 2029
Peking University Beijing, China
Bachelor, Data Science and Big Data Technology July 2024
Bachelor, Economics (the second degree) July 2024
University of California, Berkeley Berkeley, CA
Exchange Student Jan. 2023 - Aug. 2023

PUBLICATIONS

[1] **Siyuan Dong**, Zhuochen Fan, Tianyu Bai, Tong Yang, Hanyu Xue, Peiqing Chen and Yuhan Wu. "M4: A Framework for Per-Flow Quantile Estimation". In 2024 IEEE 40th International Conference on Data Engineering (ICDE).

[2] Yuhan Wu, **Siyuan Dong (Co-first)**, Yi Zhou, Yikai Zhao, Fangcheng Fu, Tong Yang, Chaoyue Niu, Fan Wu, and Bin Cui. "Kvsagg: Secure aggregation of distributed key-value sets". In 2023 IEEE 39th International Conference on Data Engineering (ICDE).

[3] Yuhan Wu, Shiqi Jiang, **Siyuan Dong (Co-first)**, Zheng Zhong, Jiale Chen, Yutong Hu, Tong Yang, Steve Uhlig, and Bin Cui. "MicroscopeSketch: Accurate Sliding Estimation Using Adaptive Zooming." In Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, pp. 2660-2671. 2023.

[4] Zhuochen Fan, Yubo Zhang, **Siyuan Dong**, Yi Zhou, Fangyi Liu, Tong Yang, Steve Uhlig, and Bin Cui. "HoppingSketch: More Accurate Temporal Membership Query and Frequency Query." IEEE Transactions on Knowledge and Data Engineering (2022).

[5] Yuhan Wu, Shiqi Jiang, Yifei Xu, **Siyuan Dong**, Kaicheng Yang, Peiqing Chen, and Tong Yang. "Unbiased Real-time Traffic Sketching." IEEE Transactions on Network Science and Engineering (2023).

WORKS IN PROGRESS

[6] **Siyuan Dong**, Junran Jia, Chiyu Hao, Zhongwei Xu, Haotian Gong, Donna Pham, Lin Ma. "An Empirical Evaluation of Pretrained LLMs for Query Plan Representation", in progress.

[7] **Siyuan Dong**, Chiyu Hao, Zhongwei Xu, Haotian Gong, Donna Pham, Lin Ma. "Towards Universal and Efficient Query Plan Representation with LLMs", in progress.

[8] **Siyuan Dong**, Yuxuan Tian, Wenhan Ma, Tong Yang, Chenye Zhang, Yuhan Wu, Kaicheng Yang and Yaojing Wang. "Detecting Flow Gaps in Data Streams", in progress.

RESEARCH EXPERIENCE

University of Michigan, Ann Arbor — Ann Arbor, MI
Research with Prof. Lin Ma — 2024 – Present

Large Language Models for Database Query Plans.

- **An Experimental Evaluation of LLMs for Query Plan Representation (QPR) [6].**
  - **Why it matters.** In ML4DB pipelines, query plans must be embedded into fixed-size vectors before conducting downstream tasks (cost/cardinality estimation, knob tuning). The quality of this representation directly bounds model accuracy and robustness.
  - **Advantage of LLM: Off-the-shelf.** *Pretrained LLMs* can serve as *zero-shot* plan encoders *without any bespoke feature design, implementation, or training*—leveraging models already trained by large tech companies. This feature also provides *forward-compatibility*: as operators/schemas evolve, new engines appear, or workloads shift, *pretrained LLMs* remain usable *without retraining or fine-tuning*. Prior QPR models, in contrast, are tightly coupled to specific databases or workloads and thus lack adaptability to new systems or evolving query plans.
  - **How we study it.** Built a controlled evaluation pipeline across *TPC-H*, *TPC-DS*, *IMDB*, and *STATS*; compared general-purpose LLMs (Bert, Llama, Gemma, Qwen) with prior SOTA QPR models on cost/cardinality estimation task.
  - **Results.** *Pretrained LLMs* consistently achieve first-tier accuracy across all datasets and tasks, demonstrating strong zero-shot capability. The primary limitation lies in their higher inference cost compared to prior QPR models.

- **Towards Universal and Efficient Query Plan Representation with LLMs [7].**
  - **Goal.** Design a *general* (no per-DB customization) and *efficient* (small model / low inference cost) LLM-based QPR model.
  - **Method.** (i) A multi-modal learning framework that jointly models semi-structured signals from the query plan side and structured signals from the data side (e.g., table statistics, histograms), enabling richer and more generalizable plan representations. (ii) Compact, schema-aware tokenization that preserves operator/edge semantics while shortening token sequences. (iii) Lightweight adapters / knowledge distillation from larger teacher LLM models into smaller student LLM models;
  - **Challenges.** Cross-DB variation in physical properties and plan lexicon; retaining generality across databases; keeping embedding quality under tight latency and memory budgets.

Peking University — Beijing, China
Research guided by Prof. Tong Yang and Prof. Bin Cui — 2021 – 2024

Data Systems, Streaming Analytics, and Privacy.

- **Per-Flow Quantile Estimation Framework [1].**
  - Converted single-flow quantile algorithms into per-flow estimation via sketch-based memory allocation and flow-size-aware treatment to balance coverage of small flows with precision for large flows.

- **Secure Aggregation of Distributed Key–Value Sets [2].**
  - Extended secure aggregation beyond vectors to sparse key–value sets for Federated Learning (FL); co-designed HyperIBLT, a homomorphic transform enabling bidirectional KV $\leftrightarrow$ vector conversion; implemented a GPU-friendly version and evaluated accuracy/throughput/memory in FL and global frequency estimation.

- **Sliding-Window Frequency Estimation and Change Detection [3].**
  - Designed compact windowed frequency representations exploiting temporal similarity; implemented and benchmarked against SOTA on frequency estimation, finding Top-k, and detecting heavy-change tasks, improving both accuracy and throughput.

- **Detecting Flow Gaps in Data Streams (variation monitoring) [8].**

- Introduced variation monitoring beyond sums/cardinality; engineered sequence-number indexing and a two-tier design (broad scan + targeted drill-down) to save memory and accelerate detection.

University of California, Berkeley Berkeley, CA
NetSys Lab, supervised by Prof. Sylvia Ratnasamy 2023 – 2024

Networked and Data-Intensive Systems.

• **Just-in-Time Data Access for Evolving Schemas.**

- Extended TSBS to study how increasing schema variety impacts DB insertion/query throughput; observed overheads from dynamic column/table creation and broader predicate spaces, informing adaptive ingestion/query planning.

• **A Federative-by-Design Platform for Community Computing.**

- Co-designed application of *Comverse* to a privacy-preserving community safety workload (federated learning with camera streams), demonstrating simplified development with user autonomy preserved.

REFERENCES

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